

Carlos Moreno Rojas

Biomedical & Electrical Engineer | R&D | Embedded Systems | Instrumentation

Santiago, Chile | camoro2002@uc.cl | linkedin | Portfolio

RESEARCH PROFILE

Biomedical and electrical engineer and M.Sc. candidate working at the interface of **biomedical instrumentation, organ perfusion, embedded systems, signal processing, and NIR fluorescence imaging**. Research focuses on translating physiological questions into measurable engineering variables, building reproducible experimental platforms, and validating complete hardware-software systems for biomedical research.

RESEARCH INTERESTS

Biomedical instrumentation; organ perfusion and preservation; liver viability assessment; ICG/NIR fluorescence; biosensors; physiological monitoring; embedded systems and data acquisition; biomedical signal and image processing; medical-device prototyping; interdisciplinary experimental engineering.

EDUCATION

Pontificia Universidad Católica de Chile 2026–2027 (expected)
M.Sc. Candidate in Biomedical Engineering – Biomedical Instrumentation Santiago, Chile

- Supervisor: **Dr. Rolando Rebolledo, Prof. Domingo Mery**.
- Thesis: *Development of an experimental platform for ex vivo liver viability assessment using ICG-NIR fluorescence*.

Pontificia Universidad Católica de Chile 2025–2026
Professional Degree in Electrical Engineering Santiago, Chile

Pontificia Universidad Católica de Chile 2021–2024
Bachelor's Degree in Engineering Sciences; Major in Biomedical Engineering, Minor in Biomedical Engineering Santiago, Chile

RESEARCH EXPERIENCE

Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo & Prof. Domingo Mery Santiago, Chile
Master's Thesis Research – Ex Vivo Liver Viability and ICG-NIR 2026–Present

- Designed and developed a custom **ex vivo rat liver perfusion platform** for normothermic perfusion, integrating pressure, flow, temperature, oxygenation, pulse damping, bubble management, and thermal control.
- Developed a modular **hydrodynamic monitoring and data-acquisition system** for pressure, flow, temperature, and vascular resistance, integrating clinical sensors, embedded electronics, and real-time acquisition software.
- Calibrated and validated measurement channels against reference instrumentation, including a **Fluke ProSim 8**, using correlation, deviation, and RMSE analysis, and implemented signal sampling and filtering strategies.
- Currently developing an **ICG near-infrared imaging methodology** to characterize spatial and temporal fluorescence kinetics and investigate quantitative parameters for assessing regional liver function and heterogeneity.

Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo Santiago, Chile
Organ Preservation and Supercooling – Borealis UC 2024

- Co-developed from concept a **supercooling-based organ preservation system**, integrating mechanical design, electronics, thermal control, fluid handling, miniaturization, and experimental testing to extend preservation time for transplantation.

Medical Image Segmentation with U-Net 2025
Pontificia Universidad Católica de Chile – Dr. Marcelo Andia Santiago, Chile

- Developed a **U-Net-based neural network** for lung segmentation in CT and MRI, applying deep learning to biomedical image analysis.

SELECTED ENGINEERING AND BIOMEDICAL PROJECTS

Candel Medical Company – Omar Espinoza Jan. 2026 – Mar. 2026
Hardware/Firmware Engineering Intern - Transcranial Electrical Stimulation (tES) Santiago, Chile

- Improved a medical-device prototype toward product-level integration through **circuit redesign, PCB debugging, firmware development, and experimental validation**.
- Worked with **ESP32 and STM32**, BLE communication, signal generation and synchronization, digital control, power reduction, and hardware-software integration for tDCS/tACS/tRNS stimulation modes.

Ciervo UC – CYBATHLON 20242024

Electrical Area Leader: Robotic Transfemoral Prosthesis; CYBATHLON

Santiago, Chile / Zürich, Switzerland

- Led electrical/electronic development for a **EMG controlled robotic transfemoral prosthesis**, including prototype design and manufacturing, system integration, and active-system development.
- Integrated electronics with control and mechanical subsystems under real operational constraints; the system was taken to the international CYBATHLON 2024 competition in Zürich.

Meldic Ltda.Santiago, Chile

Electrical Engineer: FPGA-Based Upgrade for Universal Testing Machine

2025–2026

- Developed an embedded processing architecture using **Zybo Z10, Vivado, and VHDL** for force, displacement, deformation, and speed-control signals in mechanical testing equipment.

Institute for Biological and Medical Engineering (IIBM UC) – Dr. Martín MirandaSantiago, Chile

Designer: 3D Design for GE Vscan Air Accessory

2023

- Designed a 3D cover/enclosure for the **GE Vscan Air** handheld ultrasound transducer using CAD-based prototyping.

TEACHING AND LEADERSHIP

Faculty of Physics, Pontificia Universidad Católica de ChileSantiago, Chile

Laboratory Teaching Assistant – FIS154

Mar.–Jun. 2024

- Managed and supervised laboratory activities, supported students, and communicated experimental procedures and technical concepts.

Biomedical Engineering Student Chapter PUC (CEIBM)Santiago, Chile

President

2024–2025

IEEE EMBS PUC ChileSantiago, Chile

Webmaster

2024

TECHNICAL SKILLS

Biomedical / Experimental: Organ perfusion, physiological monitoring, ICG fluorescence, biosensors, medical-device prototyping, experimental design, validation, calibration, and experimental troubleshooting.

Imaging / Computer Vision: Advanced image processing, computer vision, pattern recognition, feature extraction, image segmentation, quantitative image analysis, and biomedical imaging.

Machine Learning: Deep learning, neural networks, image classification, object detection, and data-driven pattern recognition.

Optics / NIR Imaging: NIR imaging, fluorescence imaging, optical filters, image formation, camera characterization, SNR, optical acquisition, and electro-optics.

Electronics / Embedded: Analog and digital electronics, ESP32, STM32, Arduino, clinical sensors, PCB design, debugging, instrumentation, and hardware-software integration.

FPGA / Digital Design: Vivado, VHDL, Zybo Z7, FPGA-based digital systems, and embedded signal-processing architectures.

Software / Data: Python, C/C++, data acquisition, signal processing, data analysis, and scientific computing.

Prototyping / System Integration: Rapid prototyping, iterative design, design-build-test cycles, electromechanical integration, sensor integration, system-level troubleshooting, and proof-of-concept development.

Design / Fabrication: Fusion 360, CAD, 3D printing, machining concepts, mechanical prototyping, and component integration.

Leadership / Collaboration: Technical leadership, interdisciplinary teamwork, project coordination, technical communication, teaching and mentoring, and collaborative problem-solving.

LANGUAGES

Spanish: Native

English: TOEIC 900/990 – B2 / ALTE 3